Aria Kidr

ariakidr@icloud.com | +44 7578793190 | Birmingham, UK | LinkedIn | GitHub | Portfolio

EDUCATION

MSci Theoretical Physics, University of Birmingham **Sep. 2024 – Jun. 2028**

Expected 2028 | Current Weighted Average: 75% (First-Class Standard).

Relevant Modules: Chaos (**81%**), Hamiltonian Mechanics (**80%**), Foundations in Data Science (**79%**).

Sandwell Academy **Sep. 2017 – Jun. 2024**

A-Level: Mathematics (**A***), Physics (**A**) and Further Mathematics (**A**).

RESEARCH EXPERIENCE

University of Birmingham **Jun. 2026 – Aug. 2026**

Summer Research Intern *Sun, Stars, and Exoplanets Group*

- Developed and optimised an autoencoder to denoise and reconstruct stellar power spectra, using controlled simulations and learning curves to refine the architecture.
- Generated physically informed synthetic spectra and processed Kepler observations for approximately 5,000 red giants into reproducible training and validation datasets.
- Analysed latent representations and investigated nonlinear recovery of the frequency of maximum oscillation power using reconstruction losses, residual diagnostics and signal-processing baselines.
- Maintained documented code on GitHub and discussed the research at the 2026 National Astronomy Meeting.

University of Birmingham **Jun. 2025 – Sep. 2025**

Summer Research Assistant *Theoretical Physics Group*

- Developed a configurable Python simulation of the 2D Ising model, implementing Hamiltonian, energy, magnetisation and local spin calculations.
- Applied the Metropolis Monte Carlo across temperatures, initial conditions and Hamiltonian specifications.
- Performed systematic parameter sweeps identifying 84 distinct low-energy configurations and analysed convergence through energy, magnetisation and lattice-evolution visualisations.

SCIENTIFIC COMPUTING AND COMPETITIONS

UKRI STFC Scientific Computing **Oct. 2026 – Dec. 2026**

Computing Insight UK (CIUK) Cluster Challenge *Manchester, UK*

- Competed in the national high-performance computing (HPC) cluster challenge, configuring Linux environments, managing job scheduling and benchmarking scientific computing workloads under strict competition constraints.
- Worked as a team of 6 on challenges given by the University of Edinburgh, Durham and University College London.

University of Birmingham **Jun. 2026**

BlueBEAR HPC Challenge *Birmingham, UK*

- Completed five team-based Python, Bash and HPC challenges on the university supercomputer, including galaxy classification, ray tracing and supercomputer network-topology design.

ADDITIONAL EXPERIENCE

MyTutor **Oct. 2024 – Present**

Tutor *Remote*

- Currently tutoring Mathematics and Physics at A-Level and GCSE, adapting explanations to individual needs.
- Have had multiple students come back to me proud of their results coming back beyond target.

DH Furniture Group Ltd **Jul. 2022 – Sep. 2022**

Furniture Assembler *Birmingham, Perry Barr*

- Operated a CNC cutting machine and assembled furniture to technical specifications and quality standards.

TECHNICAL SKILLS AND ADDITIONAL INFORMATION

- **Programming:** Python; NumPy, Pandas, SciPy, Matplotlib, TensorFlow/Keras, Jupyter, Git/GitHub.
- **Awards & Activities:** Birmingham Project Award for developing a proposal integrating Siemens technologies into sustainable campus education.
- **Languages:** English, Kurdish (Sorani) | **Interests:** Swimming, Badminton and Film Criticism.